

HW03 — GUI & Usability Testing on EMS

Course: CS423 / CSC15003 — Software Testing (AI-augmented · 2026) **Student:** Huỳnh Gia Ân · MSSV 23127153 · hgau23@clc.fitus.edu.vn · Nhóm 2 **System under test:** HCMUS EMS — <https://prod-dev.ems-fitus.cloud/>
Scenario: D — Support request lifecycle (user submits a request, admin resolves it) **Repository (public):** <https://github.com/huynhgiaau27112005/23127153-hw03-ems> (branch hw03-submission) **Report date:** 2026-08-06

1. Scope: scenario and screens

Scenario **D** covers the full support-request lifecycle, which is the only EMS flow that crosses the user/admin boundary while still fitting in three screens. That makes it possible to test the same domain object (a support request) as it is *created*, *tracked*, and *resolved*, so consistency defects between the user side and the admin side become visible instead of hiding inside one role.

ID	Screen	URL	Role	Why this screen
D1	Create support request	/complaints/new	Student	The only data-entry form in the scenario: labels, required-field marking, validation, error recovery and file upload all live here (IA-02 coverage).
D2	My support requests (list + detail)	/complaints , /complaints/97	Student	List/filter/empty-state and read-only detail: covers navigation, status feedback and list-level empty/loading states (IA-01, IA-03, IA-04).
D3	Admin support request management (list + detail)	/dashboard/admin/complaints , /dashboard/admin/complaints/97	Admin	The resolution side: status tabs, multi-filter search, date range, respond form. Lets the same request (#97) be compared across roles.

Accounts used

Role	Account
Student (D1, D2)	23127153@student.hcmus.edu.vn — ÂU HUỖNH GIA
Admin (D3)	admin@gmail.com

Seed data: request #97 — “HW03 test - registration failed”, category *Support*, status *Pending*, created by ÂU HUỖNH GIA · 23127153. The same request is used as the anchor object on D2 and D3.

2. Task 1A — Shared GUI checklist (group artefact)

Item	Value
Checklist file	group/GUI_Checklist.md

Item	Value
Aspects	IA-01 general UI standards · IA-02 forms · IA-03 navigation · IA-04 feedback/state
Executable items	40 (IA01–01...IA04–10); the four IA*-EX worked examples are excluded from the execution count
Reference sources	group/Reference_Sources.md (Nielsen 10 heuristics, Norman's 6 principles, Shneiderman's 8 golden rules, WCAG 2.2, ISTQB FL, course slides)
AI prompts used to build it	group/Checklist_AI_Prompts.md

Each checklist row carries an explicit *Criterion* (the pass/fail condition), an *Expected* user outcome, the *Source* heuristic, and an *Origin* column that states honestly whether the row was AI-generated, AI-drafted then human-refined, or human-refined because the AI missed it (i18n, accessibility/contrast, colour-only status, double-submit were the recurring AI blind spots).

Group scenario split (Nhóm 2)

Member	Scenario
Luân (23127414) & Đỗ Trí (23127541)	A — Event administration (shared pool, approved by lecturer)
Hà (23127044)	B — Participant experience
Lê Minh Trí	C — User administration
Huỳnh Gia Ân (23127153)	D — Support requests (D1, D2, D3)

3. Task 1B — Checklist execution per screen

The 40-item checklist was executed once per screen on the live EMS (macOS · Cursor Browser · desktop viewport), giving **120 item-executions**. Every `Fail` has a note and a screenshot; every `N/A` has a stated reason (the criterion has no corresponding component on that screen).

Screen	Pass	Fail	N/A	Total	Detailed matrix
D1 — Create support request	23	6	11	40	evidence/t1b/D1_execution.md
D2 — My requests (list + detail)	26	4	10	40	evidence/t1b/D2_execution.md
D3 — Admin support management	25	4	11	40	evidence/t1b/D3_execution.md
Total	74	14	32	120	

3.1 D1 — Create support request (23 / 6 / 11)

Checklist ID	Defect observed	Screenshot
IA01-06	<code>document.title</code> is Vietnamese (Gửi yêu cầu hỗ trợ HCMUS EMS) while the H1 and all field labels stay English — two locales in one view.	evidence/t1b/D1/IA01-06.png
IA02-06	Submitting an empty form produces one combined alert at the bottom of the page listing every missing field, instead of inline messages next to each field.	evidence/t1b/D1/IA02-06.png

Checklist ID	Defect observed	Screenshot
IA02-07	After the failed submit, focus stays on the Submit button rather than moving to the first invalid field.	evidence/t1b/D1/IA02-07.png
IA03-01	On /complaints/new the menu entry “Support requests” has no active/current state.	evidence/t1b/D1/IA03-01.png
IA03-03	No breadcrumb on the create flow.	evidence/t1b/D1/IA03-03.png
IA03-09	The floating social FAB overlaps the content edge near the form actions.	evidence/t1b/D1/IA03-09.png

Note on IA02-02: initially scored Fail, then **re-scored Pass** — the required-field asterisks are rendered through CSS (::after) and are visible in the UI even though they are not in the DOM text.

3.2 D2 — My support requests (26 / 4 / 10)

Checklist ID	Defect observed	Screenshot
IA01-06	List page: document.title = Yêu cầu hỗ trợ HCMUS EMS (VI) while H1 = “Support requests” (EN).	evidence/t1b/D2/IA01-06.png
IA03-01	On the detail page /complaints/97 the “Support requests” menu item still has no clear current state.	evidence/t1b/D2/IA03-01.png
IA03-03	No breadcrumb on either the list or the detail view.	evidence/t1b/D2/IA03-03.png
IA03-09	Social FAB sits over list/detail content.	evidence/t1b/D2/IA03-09.png

3.3 D3 — Admin support request management (25 / 4 / 11)

Checklist ID	Defect observed	Screenshot
IA01-06	With the UI switched to Vietnamese, document.title stays English (Support Request Management HCMUS EMS) and the admin chrome keeps English strings (Switch language, Back to user dashboard, Notifications, Go to page); the detail page also keeps an English “Internal note” heading.	evidence/t1b/D3/IA01-06.png
IA02-02	The admin respond form marks neither required nor optional fields and gives no legend for “Nội dung phản hồi” / “Internal note”.	evidence/t1b/D3/IA02-02.png
IA02-08	The date filter accepts From 2026-12-31 > To 2026-01-01 without any validation or error; the user only sees an empty result list.	evidence/t1b/D3/IA02-08.png
IA03-03	No breadcrumb on the admin list or detail.	evidence/t1b/D3/IA03-03.png

3.4 Cross-screen pattern

Three defects reproduce on all three screens — mixed VI/EN localisation (IA01-06) and the missing breadcrumb (IA03-03) everywhere, plus the social FAB and the weak nav active-state on both student screens. These are

therefore reported as *consolidated* findings in the Findings Log rather than as one finding per screen, with all affected screens and screenshots listed on the single row.

4. Task 2 — Usability Report (5 real users)

Full raw artefacts: `evidence/t2/task_scenario.md` , `participants.md` , `sessions_P1-P5.md` , `sus_scores.md` , `metrics.md` . Method: facilitated structured walkthroughs on the live EMS (facilitator Huỳnh Gia Âu), not unmoderated remote tests.

4.1 Participants

P	Name	Profile	Contact (masked)
P1	Huỳnh Gia Duy	Student	097***777
P2	Huỳnh Gia Bảo	Student	093***222
P3	Huỳnh Tích Đức	Guest	097***777 (same household as P1 — recorded as-is)
P4	Trần Thị Thảo	Guest	091***493
P5	Nguyễn Phúc Ti Na	Guest	098***367

4.2 Aggregate metrics

Metric	Value
Task success (Completed)	3 / 5 (60%) — P1, P2, P4
Partial	2 / 5 — P3, P5
Mean / median time on task	368.6 s / 365 s
Mean errors per session	3.8
Mean SUS	55.0 (below the ~68 average)

Per-participant SUS: P1 75.0 · P2 60.0 · P3 42.5 · P4 62.5 · P5 35.0.

4.3 Ranked usability findings (Nielsen severity 0–4)

Sev	ID	Title	Screens
3	U-001	Combined required-field alert at page bottom; focus stays on Submit	D1
3	U-005	Admin date filter accepts From > To	D3
2	U-002	Mixed VI/EN localisation	D1, D2, D3
2	U-004	Missing breadcrumb	D1, D2, D3
2	U-003	Social FAB overlaps form/list chrome	D1, D2

Evidence screenshots: `evidence/t2/U-001_...png` ... `U-005_...png` . Recommendations (P0: inline validation + date-range check; P1: locale + breadcrumb; P2: FAB) are in `evidence/t2/metrics.md` .

5. Task 3 — Cross-browser / cross-platform report

Full matrix: `evidence/t3/matrix.md` . Runbook for real-device follow-up: `evidence/t3/browserstack_runbook.md` .

Method (honest): 15 cells (5 per screen × D1/D2/D3) were captured with local Chrome CDP

`Emulation.setDeviceMetricsOverride` + UA override, each PNG carrying the overlay

`23127153@clc.fitus.edu.vn` . Every cell is labelled **local browser + device-metrics emulation (not a real device)**.

BrowserStack Live real-device recapture is pending student login (tab opened; see runbook). Coverage rule is satisfied per screen for OS ∈ {Windows, macOS, Android}, browser ∈ {Chrome, Firefox, Safari, Edge, Samsung Internet}, device class ∈ {desktop, tablet, phone}.

Screen	Pass	Fail	Not executed
D1	5	0	0
D2	5	0	0
D3	4	1 (C3 phone overflow)	0
Total	14	1	0

XP-001 (Major): D3 admin list horizontal overflow at phone width (`scrollWidth` 516 > `clientWidth` 412) — `evidence/t3/D3/C3.png` .

6. Bug & Usability Findings Log

Canonical file: `Bug_Usability_Findings_Log.md` .

Family	Count	Severity mix
F-* (Task 1B, consolidated from 14 Fail rows)	7	3 Major · 4 Minor
U-* (user testing)	5	2 × sev-3 · 3 × sev-2
XP-* (cross-platform)	1	1 Major
Total	13	

Google Form channel: the lecturer waived the Google-Form submission for this cohort, so every finding row carries the timestamp value N/A – form waived by lecturer . The aggregated log is therefore the single authoritative channel for this submission.

7. Agent Skills

Three reusable skills were authored so the same procedure can be replayed on any other EMS screen:

Skill	Purpose	Path
<code>ems-gui-checklist</code>	Execute the 40-item GUI checklist on one screen, enforcing Pass/Fail/N/A scoring and the <code>evidence/t1b/{screen}/{ID}.png</code> Fail-evidence rule.	<code>skills/ems-gui-checklist/SKILL.md</code>

Skill	Purpose	Path
ems-usability-session	Run and record one facilitated usability session: scenario, observation fields, SUS, probes.	skills/ems-usability-session/SKILL.md
ems-compat-matrix	Plan and execute the cross-platform matrix with the OS × browser × device-class coverage rule and the student-ID overlay.	skills/ems-compat-matrix/SKILL.md

The skills were used for real: D1, D2 and D3 in §3 were each executed through `ems-gui-checklist`, which is why the three execution files share an identical structure and the same evidence-path convention.

Demo video: pending student recording/upload — script at `Demo_Recording_Script.md`. Until the YouTube URL is filled in, the Agent Skills self-grade is reduced (see §10).

8. AI Audit and AI Critique

- `AI_Audit_Report.md` — declaration, tool list, and the per-interaction log (tool, date/time, prompt, output summary).
- `AI_Critique.md` — 200–300 word critique of where the AI was wrong, biased or incomplete during this assignment.

9. Git commit log

Each testing step is its own commit; the full log is exported to `git_commit_log.txt`.

Commit	Date (ISO)	Step
dfd2800	2026-08-05 21:50	design spec
4364f0d	2026-08-05 21:52	implementation plan
d8b1372	2026-08-05 21:57	shared GUI checklist IA-01...04 + group references
003a4d1	2026-08-05 22:12	EMS GUI/usability/compat agent skills
9b4663a	2026-08-05 22:28	SUT URL corrected to <code>prod-dev.ems-fitus.cloud</code>
02ab93f	2026-08-05 23:01	checklist execution — D1
01cb182	2026-08-05 23:09	D1 count/evidence correction
857f862	2026-08-05 23:25	D1 Fail screenshots realigned, IA02-02 re-scored
c5e83b0	2026-08-05 23:39	checklist execution — D2
e8a3d67	2026-08-05 23:47	D2 counts + list i18n evidence corrected
2ed4270	2026-08-06 00:06	checklist execution — D3
fc0836c	2026-08-06	AI Audit Report + AI Critique (+ PDF)
a6e449b	2026-08-06	demo recording script + packaging verifier
(subsequent)	2026-08-06	findings log · 5-user sessions · cross-platform matrix · report finalisation

10. Self-assessment

Target overall ~**085**. Honest deductions: BrowserStack real-device cells not yet substituted for local emulation (Task 3), and Agent Skills demo URL not yet uploaded (Task 5).

No.	Criteria	Max	Self
1a	Task 1A — Shared checklist (>40 items) + refs + AI prompts	15	14
1b	Task 1B — Execution on ≥3 screens + bug reports	15	14
2	Task 2 — 5-user testing → Usability Report	25	22
3	Task 3 — Cross-browser / cross-platform matrix	25	18
4	Findings Form + aggregated log	10	9
5	Agent Skills (+ demo)	10	7
	Total	100	084

Notes: (3) matrix coverage rule satisfied with honest local CDP emulation + runbook; self-score held back until BrowserStack Live replaces the cells. (5) three skills exist and were used on D1–D3; demo YouTube URL still pending — script ready. (4) Form waived by lecturer; log is complete and consistent.